# Techfest 2026-27

इलेक्ट्रॉनिकी एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
**ELECTRONICS AND INFORMATION TECHNOLOGY**

# UAV-X: Resilient BVLOS Swarm Challenge

## Overview:

PUSHPAK Grand Challenge 2026: Towards Drone Excellence

A National Innovation Challenge under the PUSHPAK: National Mission on Drone Technology.

Funded by the **Ministry of Electronics and Information Technology (MeitY), Government of India.**

Hosted by **IIT Bombay.**

In collaboration with **IISER Bhopal, Indian Institute of Science Education and Research Bhopal, Madhya Pradesh (IISERB)**

India is entering a new era of autonomous aerial systems, intelligent mobility, and drone-enabled digital infrastructure. The PUSHPAK Grand Challenge 2026 is a national platform created to identify and accelerate high-impact engineering solutions that can address real-world problems through drones, AI, robotics, sensing, communication systems, and autonomous technologies.

This is not a conventional hackathon or idea competition. The Grand Challenge is designed to take promising technical concepts through evaluation, mentoring, prototyping, testing, and deployment-oriented development. Selected teams will work with researchers, domain experts, and industry mentors to transform innovative ideas into functional, field-ready prototypes.

Grand Challenge 1, presented by IISERB at IIT Bombay Techfest 2026-27, is titled "UAV-X: Resilient BVLOS Swarm Challenge."

## Guiding Professors / Mentors

Grand Challenge 1 (UAV-X: Resilient BVLOS Swarm Challenge) will be guided by the following faculty mentors from the collaborating institutes:

**Prof. Arnab Maity**

*IIT Bombay – Aerospace Engineering*

Prof. Arnab Maity is an accomplished aerospace researcher and faculty member at IIT Bombay with strong expertise in dynamics, guidance, navigation and control of aerospace and autonomous systems. His key strength lies in combining rigorous theoretical foundations in control, estimation, optimization and system dynamics with their application to complex real-world aerospace problems. His research encompasses autonomous flight systems, UAV and drone technologies, guidance and navigation algorithms, adaptive and optimal control, and aerospace vehicle and propulsion-system control. With a strong analytical and systems-oriented approach, Prof. Maity contributes to the development of robust, intelligent and reliable autonomous platforms, particularly for applications where safety, precision and real-time decision-making are critical. His expertise makes him well positioned to contribute to advanced drone technology initiatives involving autonomous navigation, flight control, collision avoidance, multi-agent systems and next-generation UAV applications, while bridging fundamental research with technology development and mission-oriented outcomes.

**Prof. P.B. Sujit**

*IISER Bhopal – Department of Electrical Engineering and Computer Science*

P.B. Sujit is a Professor in the Department of Electrical Engineering and Computer Science at IISER Bhopal. His research interests span multi-robot systems, autonomy across ground, aerial and ocean going vehicles, and human-swarm interaction. He is IEEE senior member and AE for several international conferences and journals.

## Objective:

Develop autonomous UAV swarms capable of maintaining resilient beyond visual line-of-sight (BVLOS) communication while executing a disaster response mission.

## What Happens After the Grand Finale?

The journey does not end with the finale.

### Prototype Development Support (Build & Test Phase)

Selected winning teams (upto 5 teams) will be invited to participate in a dedicated Prototype Development Support Programme, beginning December 2026 onwards. This post-challenge phase is specifically designed to bridge the gap between academic/start-ups innovation and real-world deployment.

### Build the Future of India's Drone Ecosystem

If you have an idea that can redefine autonomous flight, intelligent sensing, aerial mobility, infrastructure automation, and the cybersecurity of drones, this is your opportunity.

**Design. Build. Test. Demonstrate.**

Join the **PUSHPAK Grand Challenge 2026** and help create the next generation of indigenous drone technologies for India and the world.

## Challenge Question:

### Mission Scenario

A major earthquake or landslide has disrupted terrestrial communication infrastructure. A Ground Control Station (GCS) is established outside the affected area and deploys a fleet of UAVs. The UAVs having limited flight time must survey designated Points of Interest (PoIs), collect imagery and situational data, and continuously relay information back to the GCS through a resilient multi-hop aerial communication network.

The swarm must autonomously:

- Survey all assigned disaster locations.
- Maintain end-to-end communication with the GCS.
- Dynamically assign relay UAVs.
- Reconfigure the network when communication degrades or UAVs fail or return to home for recharging.
- Prioritize newly emerging high-priority regions.
- Complete the mission safely and within the allotted time.

## Format of Competition:

### Stage 1: Preliminary Design Verification

Teams submit:

- 6–8 page technical proposal
- Software architecture
- Working proof-of-concept simulation
- Source code
- Installation instructions
- Demonstration video

Evaluation focuses on feasibility, novelty, reproducibility, and preliminary communication-aware autonomy. Approximately ~15 teams qualify.

### Stage 2: Technical Challenge

Qualified teams receive advanced disaster scenarios with hidden disturbances such as UAV failures, communication outages, packet loss, and new emergency tasks.

Teams submit:

- Final source code
- Technical report
- Simulation logs
- Reproducibility instructions
- Demonstration video

The top 10 teams qualify for the Grand Finale.

### Stage 3: Grand Finale

Teams present their approach, answer technical questions, and execute a previously unseen disaster-response scenario before the jury.

### Evaluation Criteria

- Mission Completion: 25%
- Communication Resilience: 25%
- Autonomous Relay & Role Management: 20%
- Fault Recovery & Swarm Reconfiguration: 15%
- Safety & Collision Avoidance: 10%
- Innovation & Technical Merit: 5%

### Performance Metrics

- Mission: completion rate, completion time, priority-weighted mission score.
- Communication: packet delivery ratio, latency, connectivity availability, communication downtime.
- Autonomy: relay reallocations, recovery time, network reconfiguration efficiency.
- Robustness: performance after UAV/link failures.
- Safety: collision count and minimum inter-UAV separation.
- No UAV should be without charge and they should not go out of geo-fenced areas.

### Simulation Framework

Participants may use any open-source simulation framework such as PX4 SITL, ArduPilot SITL, Gazebo, AirSim, Isaac Sim, ROS/ROS 2, Webots, CoppeliaSim, ns-3, or equivalent.

The organizers will provide mission specifications, benchmark scenarios, communication assumptions, evaluation metrics, and a standardized log format.

## Team Specification and Eligibility:

Expected Target Audience (Upto 5 members per team)

The simulation-first format keeps the barrier to entry low and enables safe, scalable participation. The challenge is aimed at:

- College students (UG and PG) from aerospace, robotics, computer science, electronics/communication, electrical, and mechanical engineering.
- Research scholars (PhD and postdoctoral) working in unmanned aerial systems, swarm robotics, wireless networking, and multi-agent systems.
- Startups and early-stage teams building drones, autonomy stacks, and communication systems seeking a pathway toward the hardware stage.
- Participants are expected to be comfortable with a robotics simulator and programming (Python/C++, ROS). Interdisciplinary teams combining controls, networking, and AI/optimization are encouraged.

**Important Eligibility Restriction**

Project staff, research staff, consultants, interns, and other personnel directly engaged with the PUSHPAK Project, and Drone Centre, or the organizing/host institutions in connection with the Grand Challenge are NOT eligible to participate in the Grand Challenge.

This restriction applies whether such personnel participate individually or as part of a team.

Any team found to include an ineligible participant may be disqualified at any stage of the Grand Challenge, including after selection or announcement of results. The organizers reserve the right to verify the eligibility of participants at any stage and their decision in this regard shall be final.

## Timeline and Key Dates:

| Stage | Timeline(s) | Participation Brief | Prize / Reward / Support |
|---|---|---|---|
| Grand Challenge Announcement | 22 August 2026 | PUSHPAK Grand Challenge 2026 opens for registrations across India. Teams of up to 5 members may register and participate. | None |

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| Stage 1: Preliminary Design Verification | Work Time: 22 August – 27 September 2026<br>Submission Deadline: On or Before 27 September 2026<br>Results Announcement: 02 October 2026 | All registered teams are eligible. Submit a 6–8 page technical proposal, software architecture, working proof-of-concept simulation, source code, installation instructions and demonstration video. (Upto 15 teams shall qualify for next stage) | INR 1 Lakh per qualifying team after the announcement of results of Stage 1 |
| Stage 2: Technical Challenge | Work Time: 03 October – 02 December 2026<br>Submission Deadline: On or Before 02 December 2026<br>Results Announcement: 06 December 2026 | Shortlisted teams receive advanced disaster scenarios with hidden disturbances and submit final source code, technical report, simulation logs, reproducibility instructions and demonstration video. (Top 10 teams qualify for the Grand Finale) | Domestic Travel & Accommodation support as per guidelines of IIT Bombay to be shared at a later stage for qualifying team |
| Stage 3: Grand Challenge Finale | 16 – 18 December 2026 | Finalist teams present and demonstrate their solution at IIT Bombay Techfest. (Upto 5 winning teams get a chance to enter the Post-Grand Challenge Stage) | Zenith – INR 3 Lakh, Horizon – INR 2.5 Lakhs, Vector – INR 2 Lakhs, Ascent (x2) – INR 1.5 Lakh, for the winning team after the announcement of results at the TechFest |

## Registration & Submission

- Participants must register for the PUSHPAK Grand Challenge on the official Techfest website: techfest.org > Competitions > PUSHPAK Grand Challenge > Choose your Challenge > Register at [www.techfest.org/competitions](www.techfest.org/competitions)

- All submissions (Stage 1 and Stage 2) are to be made via email to [pushpak_gc2026@aero.iitb.ac.in]; no submission links are used; every stage is handled entirely through email correspondence.

## Prize Money

Stage 1 (Preliminary Design Verification): INR 1 Lakh per qualifying team after the announcement of results of Stage 1.

Stage 2 (Technical Challenge): Domestic Travel & Accommodation support as per guidelines of IIT Bombay to be shared at a later stage for qualifying teams.

Stage 3 (Grand Challenge Finale):

Zenith: INR 3 Lakh

Horizon: INR 2.5 Lakhs

Vector: INR 2 Lakhs

Ascent (x2): INR 1.5 Lakh

for winning teams after the announcement of results at the Techfest.

*Note: At the end of the challenge, up to 5 winning teams will be selected. These teams will be considered for further support, funding, mentoring, and access to resources for building and testing their designs in the next phase.*

The process regarding the disbursement of prize monetty will be communicated at a later stage. Participating teams are requested to regularly check the official website and their registered email accounts for updates.

### Disclaimer

**Organizer's Rights:** The organizers reserve the right to modify, extend, postpone, suspend, or cancel the Grand Challenge, or any stage thereof, at their discretion. Any such changes will be communicated through the official communication channels of the Grand Challenge.

**Eligibility & Participation:** Participation in the Grand Challenge is subject to the eligibility criteria, rules, guidelines, and submission requirements specified by the organizers. Submission of an entry does not guarantee selection for any subsequent stage or award.

**Evaluation:** All submissions will be evaluated by the designated evaluation committee based on the criteria specified for the respective stage. The decision of the evaluation committee shall be final and binding, and no correspondence regarding the evaluation or ranking will be entertained.

**Technical Responsibility:** Participants are solely responsible for the accuracy, completeness, originality, technical feasibility, safety, and performance of their submissions. Selection or recognition of a submission by the organizers does not constitute certification, approval, or endorsement of the proposed technology for operational deployment.

**Safety & Compliance:** Participants shall ensure that their proposed solutions and demonstrations comply with all applicable laws, regulations, standards, aviation requirements, safety protocols, and permissions applicable in India. Participants shall be responsible for obtaining any permissions, approvals, or clearances required for testing or demonstration.

**Intellectual Property:** Participants shall ensure that their submissions do not infringe upon any third-party intellectual property rights. Participants retain ownership of their intellectual property unless otherwise specifically agreed under the terms of the Grand Challenge. Any use, disclosure, licensing, or transfer of intellectual property arising from the Grand Challenge shall be governed by the applicable terms and agreements communicated by the organizers.

**Confidentiality:** Participants should clearly identify any information submitted that they consider confidential or proprietary. Participants are responsible for ensuring that they do not disclose information belonging to third parties without appropriate authorization.

**Prizes & Recognition:** Prizes, awards, certificates, or other forms of recognition are subject to fulfillment of the applicable eligibility and evaluation requirements. The organizers reserve the right to withhold or modify an award where no submission meets the required standards.

**Expenses:** Unless expressly stated otherwise, participation in the Grand Challenge, including development, testing, travel, accommodation, equipment, and demonstration-related expenses, shall be borne by the participating teams.

**Liability:** The organizers, host institutions, funding agencies, partners, mentors, evaluators, and their respective personnel shall not be liable for any loss, damage, injury, delay, technical failure, or other consequence arising from participation in the Grand Challenge, except to the extent required by applicable law.

**Force Majeure:** The organizers shall not be responsible for any delay, modification, suspension, or cancellation resulting from circumstances beyond their reasonable control, including natural disasters, technical failures, regulatory changes, government directives, or other unforeseen events.

**Acceptance of Terms:** By registering for or participating in the Grand Challenge, participants acknowledge that they have read, understood, and agreed to abide by the applicable rules, guidelines, terms, and conditions of the Grand Challenge.